Experiment 3
Optical Signal Processing

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1. Introduction

In recent decades, enormous progress has been made in the fields of optical signal transmission in optical fibers and optical data storage on CD, DVD and Blu-Ray discs. In addition to these now established applications of optical technologies in information technology, efforts are also being made today to use optical processes in signal processing. This concerns, for example, the use of purely optical technologies for the coupling and decoupling of specific wavelengths in a WDM fiber ring[1] via so-called reconfigurable optical add-drop multiplexers (ROADM[2]). In the field of optical computers (optically addressable memory, optical comparator, optical flip-flop), however, the freedom of interaction of light beams penetrating through poses a problem. Elements must therefore be developed specifically for this purpose in which light can be controlled by light, i.e. a forced interaction takes place. The realization of optical signal processing is therefore a central challenge, and suitable components are being developed, manufactured and optimized in numerous laboratories. An important milestone on this path was the discovery of optical bistability in suitable semiconductor layers at room temperature and the proof that only low energies are required for the switching process. In the meantime, numerous concepts for optical switching and storage cells have been developed and presented.

The aim of this practical experiment is to understand optical signal processing using the example of a hybrid memory cell. To this end, the non-linear characteristics of an optical switching element are first investigated experimentally and compared with theoretical considerations. Based on these considerations, an optically addressable memory, an optical Schmitt trigger and optical flip-flop are then realized with these switching cells and examined with regard to signal processing properties.

NOTE:

The tasks and questions listed in this document must be completed before the experiment is carried out.

Furthermore, find out about the optical and electrical properties of photodiodes, phototransistors and luminescent diodes as well as the meaning of the terms "feedback", "non-linear system" and "time-invariant system", see e.g. references [1-4].

[1] https://en.wikipedia.org/wiki/Wavelength-division_multiplexing
[2] https://en.wikipedia.org/wiki/Reconfigurable_optical_add-drop_multiplexer

2. Optical bistability as the basis for optical signal processing

In Fig. 1a shows schematically what is meant by optical bistability. It is characterized by the hysteresis-like relation between the input and output signals of an optical element. Two things can be recognized:

- Depending on the history, there are two different values for the optical output power P_{aus} in the bistable range with a fixed optical input power P_e
- It is possible to switch between the two states, e.g. using a signal beam with the optical power P_s

An optically bistable element therefore has the properties of an optical memory and an optical switch.

Optically bistable elements can have very different characteristic curves that are customized for special applications. As an example Fig. 1b schematically shows a relationship between optical input and output power, in which P_{aus} only assumes large values when both input signals P_e and P_s are applied simultaneously. You can see the basic properties of a logical AND operation. Other logical operations can also be realized by setting suitable characteristic curves and combining them. Elements with the properties from Fig. 1 are therefore generally suitable for the construction of digital logic - e.g. for optical computers.

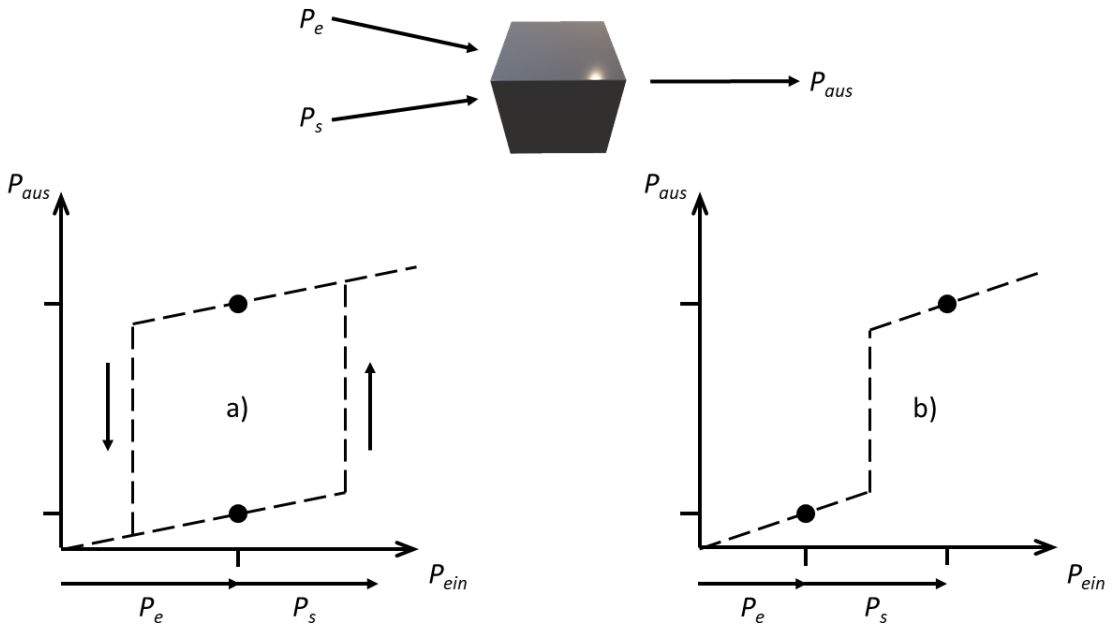


Fig. 1: Optical bistability (P is the optical power)
a) Transmission characteristic of a bistable element
b) Characteristics of a logical AND gate

The characteristic curves in Fig. 1 indicate non-linear optical elements, as P_{aus} is by no means linearly dependent on P_{ein} ! The current aim of much research work is therefore to develop materials and components with large optical non-linearities, with organic materials (polymers) playing an important role alongside semiconductor structures.

However, non-linear and bistable P_{aus} -P_{ein} characteristics can also be achieved using hybrid arrangements, as shown in Fig. 2 is shown. These optoelectronic solutions lead to an artificial optical non-linearity. In Fig. 2a shows a circuit with an electro-optical modulator, whose optical output power P_{aus} is incident on a photodetector via a beam splitter and a mirror. The photocurrent generated here controls the modulator via an electrical circuit. With sufficiently strong non-linearity and feedback, bistability as shown in Fig. 1 can be achieved.

The circuit in Fig. 2b uses a laser diode (LD) as the active element. Both the optical input signal P_e and the output signal, which is partially fed back by the beam splitter and mirror, are applied to a photodetector. This feedback can in turn generate bistability, whereby even $P_{aus} > P_{e1} + P_{e2}$ can be generated.

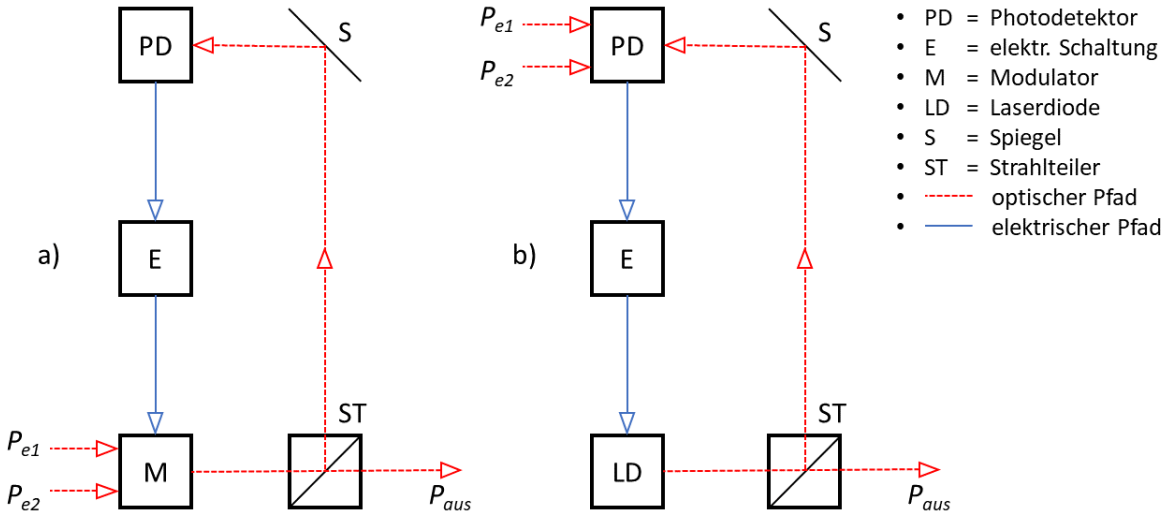


Fig. 2 Concepts of optoelectronic bistability with a) passive and b) active optoelectronic elements

The advantages of the concepts in Fig. 2 result from the fact that available electronic and optoelectronic components can be used and connected together in hybrid form. The integration of such hybrid circuits results in so-called OEICs (optoelectronic integrated circuits).

Fig. 2b also includes a very simple special case: Without the use of optical feedback via [ST] and [S], signal processing can be achieved using a suitable electronic component [E] in such a way that the optoelectronic components at the input and output merely assume the function of converters.

The circuits used in this experiment belong to this class.

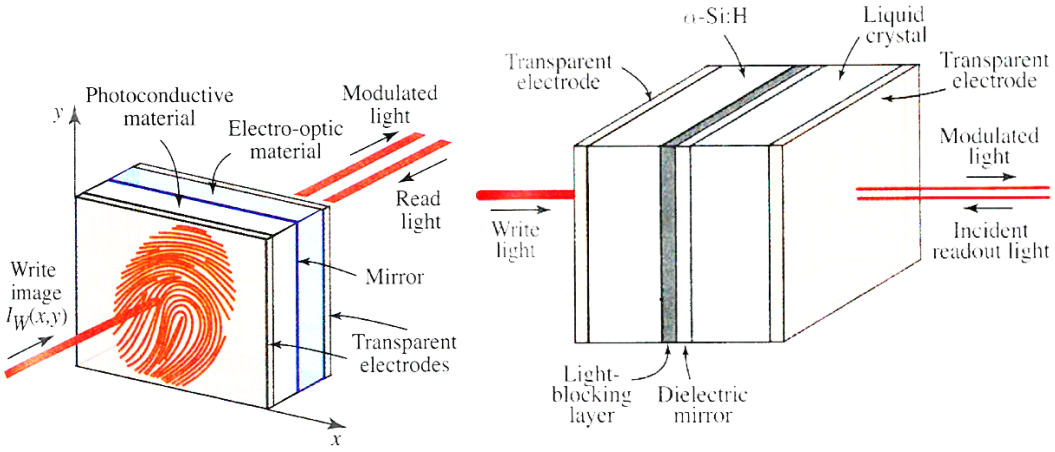


Fig. 3Optical signal processing using optically addressable SLM
(Source: B. E. A. Saleh, M. C. Teich, Fundamentals of Photonics, Wiley)

Such photonic switching and storage elements are used, for example, in two-dimensional arrays, so-called 2-dimensional spatial light modulators (SLM). [3]

In this way, parallel optical signal processing can be elegantly implemented using optically addressable SLM, as shown in Fig. 3 can be realized. A configuration of 1000 x 1000 cells, for example, leads to a data rate of 1 terabit/s at a clock frequency of 1 MHz. This advantage of optical signal processing opens up interesting areas of application in the field of image processing, pattern recognition and artificial neural networks.

[3] https://en.wikipedia.org/wiki/Spatial_light_modulator

3. The feedback optocoupler

At this point, the principle of an optical switching cell will be explained and verified in an experiment using a very clear example. In Fig. 4 shows an electrically cleverly connected optocoupler[4] whose feedback circuit consists partly of an optical and partly of an electrical signal path.

The electrical signal path is characterized by the current flow through the series resistors R_1 , R_2 , R_3 and R_V as well as by the parallel current paths along the *LED*[5] and phototransistor *PT*[6] . The optical signal path is described by the generated optical output power of the *LED*, which is received by the phototransistor. This results in an additional photocurrent within the optocoupler when the *LED is* emitting, which leads to an increase in the total current.

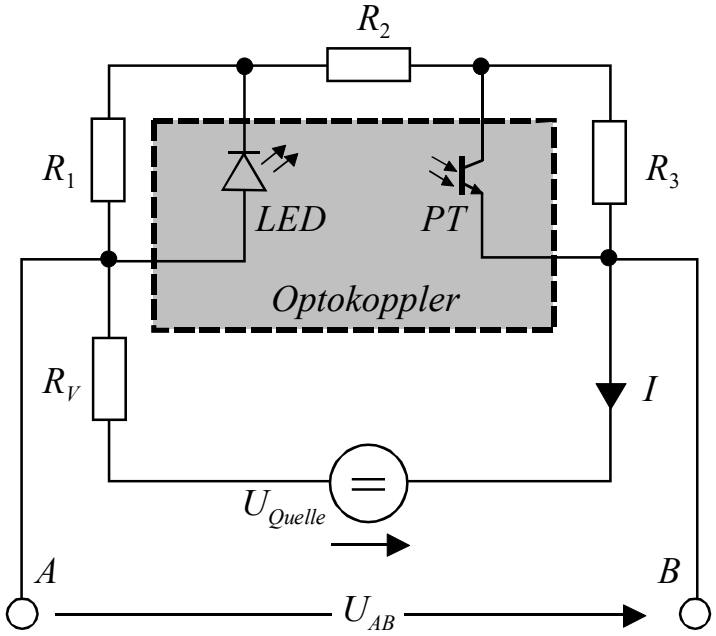


Fig. 4Circuit of the feedback optocoupler.

By varying the voltage U_{source} and neglecting the series resistor (R_V = 0), a hysteresis-like electrical characteristic is obtained with the switching processes (1) and (2) that can be observed when the voltage is impressed, see . 5.

At a certain value for the series resistor $R_V \neq 0$, the behavior changes: Instead of the hysteresis-like curve, an S-shaped curve now results, which is characterized by the dashed path (3) between the two original jump points (1) and (2) in Fig . 5 is characterized.

[4] https://en.wikipedia.org/wiki/Opto-isolator
[5] https://en.wikipedia.org/wiki/LED_circuit
[6] https://en.wikipedia.org/wiki/Photodiode#Other_modes_of_operation

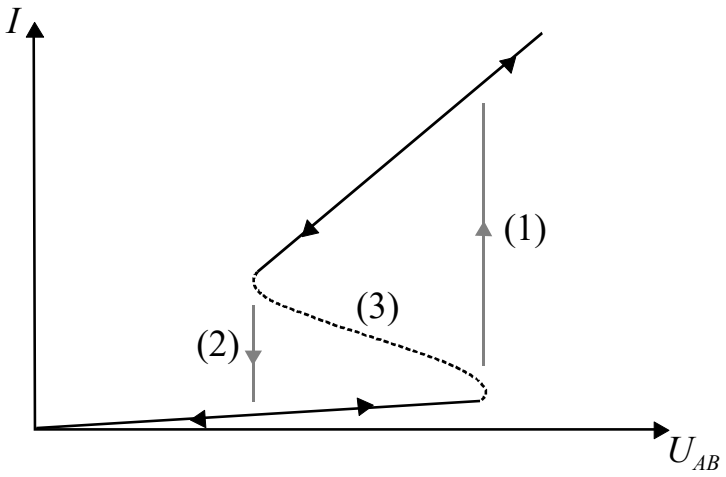


Fig . 5I-V characteristic of the optical switching element from Fig. 4 for different R_v

For a better understanding of the effect of changing the resistors R_1 , R_2 , R_3 or R_v , a simulation of the circuit of the feedback optocoupler in Spice is included in the Moodle course.

By definition, the combination of the output signal with the input signal is referred to as "feedback". There are two types of feedback: positive (positive feedback) and negative (negative feedback).

Task 1:

a) Carry out your own simulations with the LTspice model. Pay attention to the changes in the current-voltage characteristic curve when varying the resistors.

- R_1 : 1.5 kΩ , variation by± 0.2 kΩ
- R_2 : 1.0 kΩ , variation by± 0.2 kΩ
- R_3 : 7.0 kΩ , variation by± 1.0 kΩ
- R_V : 1.0 mΩ , variation by + 1.0 kΩ

b) Draw a block diagram showing positive feedback. Determine the gain factor *ACL* of the closed circuit as a function of :

- *AOL* (open loop amplification factor) and
- ß (feedback factor).

c) Look at the circuit in Fig. 6. Draw the current-voltage characteristics (I_C via V_{CE}) for different base currents I_B . Then determine the characteristic curve for $R = R_{out} + R_E$ as a function of I_C and V_{CE} . Determine the formulas for calculating the open-circuit voltage $V_{CE,OC}$ and short-circuit current $I_{C,SC}$ as a function of V_B , R_E and R_{out} .

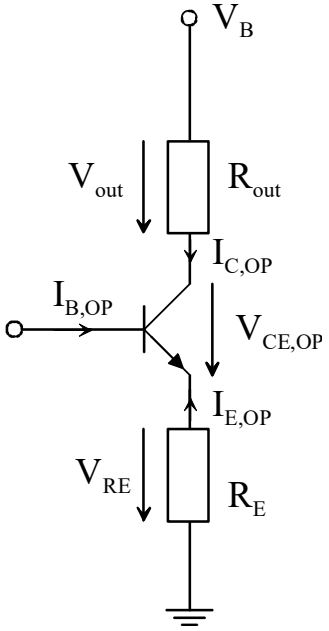


Fig. 6NPN transistor, viewed at the operating point

Question 1:

a) What is the so-called forward voltage of an LED?

b) What is the so-called current-transfer ratio in percent for an optocoupler with a phototransistor compared to one with a photodiode?

Question 2 (see LT-Spice simulation):

a) How does the characteristic curve of the feedback optocoupler come about?

b) Which components characterize the characteristic curve ranges (1) and (2)?

c) What happens at the jump points?

d) What influence do the resistors R_1 , R_2 , R_3 and the series resistor R_V have on the curve behavior? Attention: The voltage U_{AB} is always measured between points (A) and (B)!

4. The optical switching element (optically addressed memory)

In this chapter, the circuit of the feedback optocoupler is extended to include a PD_2 photodetector and an LED_2 luminescent diode (see Fig. 7). The two components are mounted in such a way that the light emitted by the LED_2 cannot reach the photodetector PD_2 - they are optically decoupled. Instead, the PD_2 photodetector is to be illuminated with the aid of another external light source (e.g. LED pocket lamp).

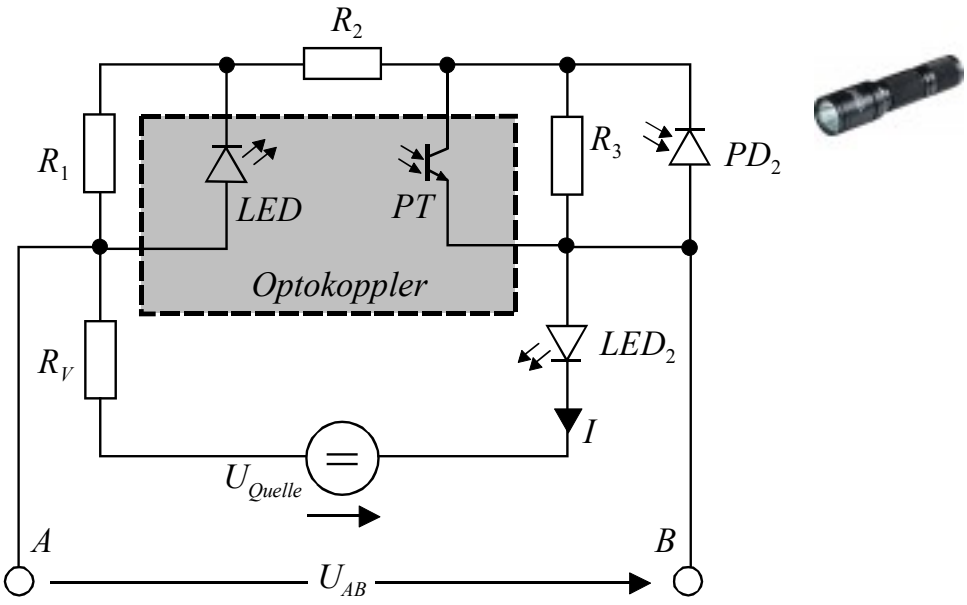


Fig. 7Extended circuit of the feedback optocoupler. (PD_2 and LED_2 are optically decoupled)

Question 3:

a) What influence (see LT-Spice simulation) does the LED_2 have on the current-voltage characteristic?

b) What is the relationship between current and optical power in a light-emitting diode and a photodiode?

c) How does illumination of PD_2 affect the feedback mechanism?

d) What is an LDR? What function does this component have when it is connected in parallel to R_1 ?

e) What is a Schmitt trigger and what can it be used for? What is the difference to a comparator?

5. The optical RS flip-flop

In the next step, two optical circuits with the same electrical and optical properties are combined to form a network - as shown in Fig. 8 shown in Fig. 8. By skillfully selecting R_V , this network can be brought into a state in which only one optical circuit can be active at a time (high-current state, see also Fig . 5). As both circuits have optical inputs and outputs and influence each other via the network, an optical RS flip-flop can be realized in this way.

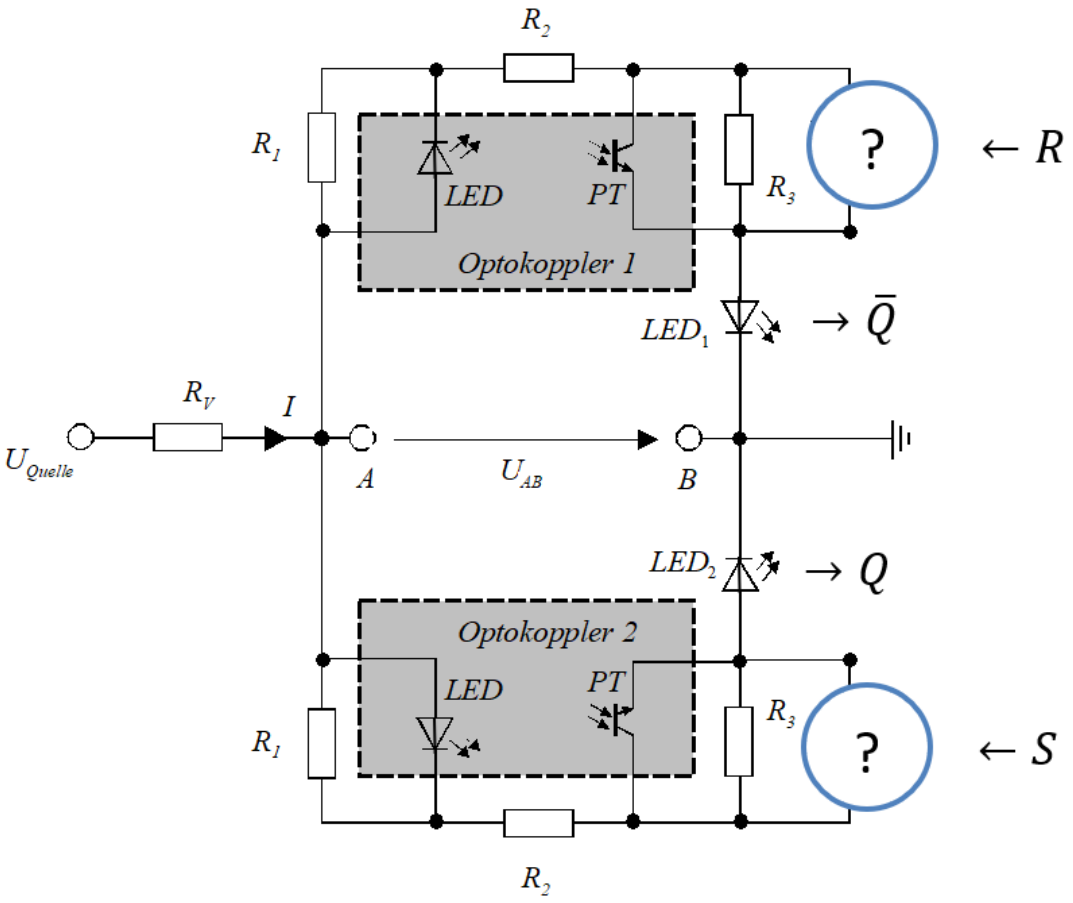


Fig. 8Network of two optical control elements interconnected to form an RS flip-flop.

Question 4:

a) Which component is hidden behind the circled question mark?

b) What could the I-V *characteristic curve of* this network look like for $R_V = 0$?

c) How must the load line for R_V be placed in the *I-V* characteristic curve *in order to* be able to switch only one switching element at a time?

6. Practical experiments

The practical experiments are divided into two sections. Firstly, the characteristics of a feedback optocoupler, the optically addressed memory and the Schmitt trigger are to be investigated. Secondly, the interconnection of two feedback optocouplers in their function as RS flip-flops is to be investigated. Pre-assembled plug-in boxes are available for carrying out the experiments. These contain the circuits to be examined, so that only the external wiring with the necessary components (function generator, DC source, oscillograph, plotter) has to be carried out.

6.1. Investigations on the optical circuit

In order to be able to make statements about the switching behavior of (opto-)electronic circuits, the evaluation of so-called current-voltage characteristics (*I-V characteristics*) is of central importance. The measurements of these characteristics for various modifications of the circuit shown in Fig. 4 optical circuit of a feedback optocoupler shown in Fig. 4 should be carried out in two different ways:

1. With the aid of a manually adjustable voltage source (i) and a flatbed plotter (ii)
2. With the aid of a function generator (iii, triangular voltage with adjustable frequency) and an oscilloscope (iv)



(i)


(iii)


(ii)



(iv)

Quasi-static tests are carried out with the first test setup (+ and - pole on the power supply unit and plotter), while the second setup allows dynamic measurements (coaxial BNC input/output on the generator and oscilloscope). To avoid any damage to the test circuits, make sure that a source voltage of U_{source} = 15 V is not exceeded.

Note: If sufficiently slow frequencies are used, a quasi-static measurement can also be carried out using the function generator!

Measure the *I-V* characteristics for various R_1 , R_2 , R_3 and R_V shown in the following table. Table 1 below. Make sure that only one resistance value is changed at a time and that other (optical) influences on the circuit are avoided. It should also be noted that the

frequency of the function generator must not be set too high (guide value for the plotter: 0.1 Hertz)!

Table 1Summary of the resistance values to be set

	R_1	R_2	R_3	R_1
Basic value	20%	50%	80%	0%
Variation	±5%	±20%	±20%	+30%

6.2. Investigations on optically addressed memory

For the following measurement, the previous circuit is to be extended to that of an optically addressed memory. A corresponding circuit diagram is shown in Fig. 9 is given. Which components must be used in parallel with the resistors R_1 and R_3 to complete the circuit?

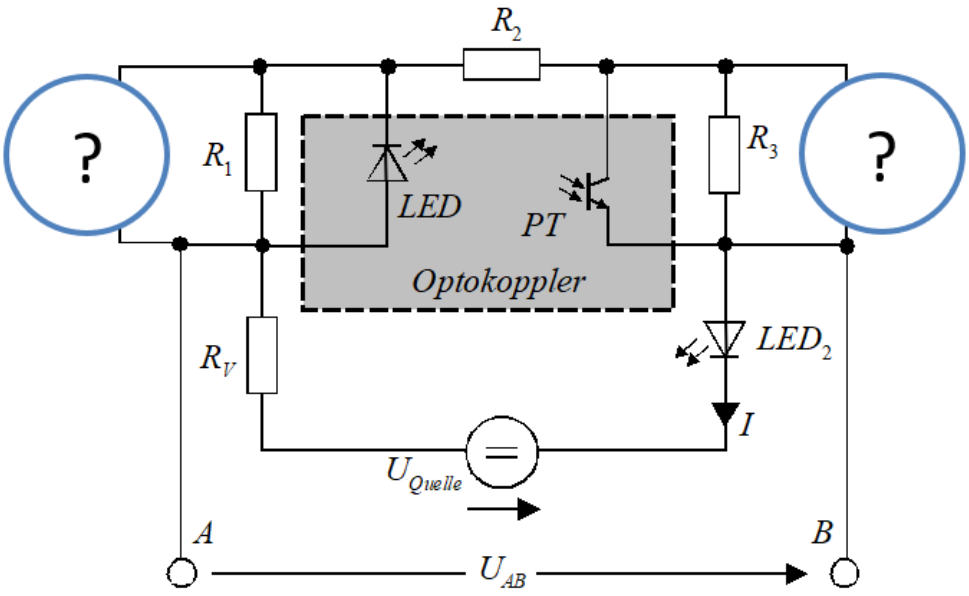


Fig. 9Circuit diagram of the optically addressed memory

Use the circuit of the optically addressed memory and use the DC voltage source to record an *I-V* characteristic for fixed values of R_1 = 20%, R_2 = 50%, R_3 = 80% and R_V = 0%. Then set a fixed voltage value that lies between the two jump points in the characteristic curve. Now illuminate the component connected in parallel to R_3 . What do you observe? What happens when the component connected in parallel to R_1 is illuminated?

6.3. Investigations on the optical Schmitt trigger

Another modification of the feedback optocoupler circuit is its configuration as a so-called Schmitt trigger. External sources must be consulted for a precise explanation of the function - also with regard to the differences to a classic comparator circuit.
As shown in Fig. 10 the circuit is additionally illuminated with a light-emitting diode (LED_3) on the component connected in parallel to R_3, which in turn is connected to a function generator and thus enables the input signal of the circuit (triangular, sine or square wave voltage) to be varied.

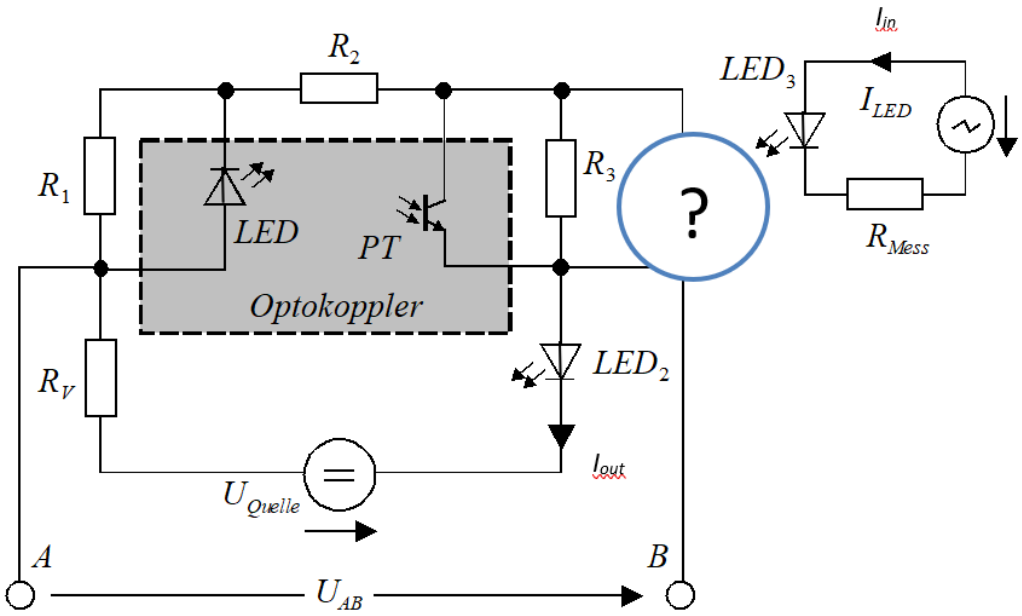


Fig. 10Circuit diagram of the optoelectronic Schmitt trigger

Set up the optoelectronic Schmitt trigger and set a DC voltage of U_{source} = 5 V. Measure the relationship between I_{out} and I_{in} at a frequency of 50 Hz (triangle voltage) on the oscilloscope in X-Y representation. Now vary the resistors R_1 , R_2 , R_3 and R_V until you see the characteristic curve of a Schmitt trigger on the oscilloscope. How does the characteristic curve change if you use different input signals at different frequencies (triangle, at 50 Hz; square wave at 50 Hz and 2 kHz)?

6.4. Investigations on the optical RS flip-flop

As already explained in the basics and shown in Fig. 8 the realization of an RS flip-flop with optoelectronic components requires the combination of two identically wired optocouplers. Setting (S) and resetting (R) should be carried out by means of optical irradiation. The resulting state of the flip-flop is to be indicated by LEDs ($LED_2 = Q$, $LED_1 = \bar{Q}$). The correct setting of the series resistor R_V is of central importance for correct functioning in terms of the truth table of a classic RS flip-flop!

Use the quasi-static (plotter) and dynamic measuring station (oscilloscope) to record the *I-V characteristic curve* when varying R_V . To estimate sensible settings for R_V , the

dynamic measuring station with oscillograph should be used first. For the other resistance values, the values given in Table 2

a) apply

Table 2: Resistance values for the investigation of the RS flip-flop

R_1	R_2	R_3
20%	50%	100%

b) Set R_V to a suitable value with which the states defined in the truth table can be achieved. Do not use the function generator for this, but the DC voltage source. Check the function of the circuit as an RS flip-flop by illuminating *S* and *R* alternately.

c) Complete the truth table for the RS flip-flop with the measurement results obtained (note the correct setting of R_V)

c.1. 0→ P_{in} = 0

c.2. 1→ P_{in} = Optical power that activates the circuit

Table 3Truth table of an RS flip-flop

		Theory		**Experiment**	
$\boldsymbol{R}$	$\boldsymbol{S}$	$\boldsymbol{\bar{Q}}$	$\boldsymbol{Q}$	$\boldsymbol{\bar{Q}}$ $(\boldsymbol{LED_1})$	$\boldsymbol{Q}$ $(\boldsymbol{LED_2})$
0	0				
1	0				
0	1				
1	1				

7. Literature

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